

Virtual Twin Approach Using Physiologically Based Pharmacokinetic Modeling to Support Precision Dosing of Valproic Acid in Geriatric Patients

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Personalized dosing is particularly important for drugs with narrow therapeutic indices in geriatric patients, who exhibit substantial physiological variability and limited pharmacokinetic (PK) evidence to guide individualized dose selection. Valproic acid (VPA) is an effective treatment option for bipolar disorder in older adults, yet dosing largely relies on therapeutic drug monitoring (TDM), which guides adjustment only after steady state is achieved. This study evaluated a physiologically based pharmacokinetic (PBPK)-guided virtual twin (VT) framework to prospectively predict individual VPA exposure in a real-world geriatric population. A PBPK model for extended-release VPA was developed using the Simcyp Simulator (version 25). Clinical data ($n = 74$) were collected from elderly patients receiving VPA at the Samsung Medical Center. PBPK model performance was validated against published PK data from healthy adults and independent TDM data from the geriatric patients. The model adequately reproduced observed PK of VPA, with predicted PK parameters ranging from 0.73- to 1.37-fold of observed values. Virtual twins were subsequently generated for each patient by incorporating patient-specific demographic characteristics and routinely available physiological parameters. Individual PK profiles were simulated and compared with measured TDM concentrations. Incorporation of demographic and physiological covariates achieved an absolute average fold error of 1.13 (90% CI: 1.10–1.16), with nearly 90% of predictions falling within the 0.8- to 1.25-fold range of the observed concentrations. These findings suggest that PBPK-guided virtual twins can predict individual VPA exposure using routinely available clinical data and may enable prospective dose optimization, complementing conventional TDM to advance precision dosing in geriatric pharmacotherapy.

Study Highlights

What is the current knowledge on the topic?

✓ Valproic acid (VPA) is a narrow therapeutic index drug widely used in geriatric patients, yet dosing remains challenging due to substantial interindividual variability and limited pharmacokinetic (PK) data in elderly populations. Physiologically based pharmacokinetic (PBPK) modeling combined with a virtual twin (VT) approach has emerged as a promising strategy to predict individual drug exposure using patient-specific clinical information.

WHAT QUESTION DID THIS STUDY ADDRESS?

✓ Can a PBPK-guided VT framework accurately predict individual VPA exposure in real-world geriatric patients and support model-informed precision dosing?

WHAT DOES THIS STUDY ADD TO OUR KNOWLEDGE?

✓ This study demonstrates that a VT model can reliably predict VPA trough concentrations in elderly patients using routinely

available demographic and physiological data. Progressive incorporation of patient-specific covariates improved predictive performance, achieving bioequivalence-level accuracy, with nearly 90% of predictions within the 0.8- to 1.25-fold range of the observed concentrations.

HOW MIGHT THIS CHANGE CLINICAL PHARMACOLOGY OR TRANSLATIONAL SCIENCE?

✓ PBPK-guided VT may enable prospective dose optimization for VPA before treatment initiation, complementing conventional TDM and reducing the risk of over- or underexposure in vulnerable geriatric patients. This approach supports implementation of model-informed precision dosing using routinely collected clinical data and provides a practical pathway toward bedside application of quantitative pharmacology in geriatric care.

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Recent advances in quantitative clinical pharmacology and systems pharmacology have accelerated the development of personalized dosing,¹ a paradigm that seeks to optimize pharmacotherapy by accounting for interindividual variability in drug response. Such variability arises from physiological, environmental, and disease-related factors and may substantially influence both pharmacokinetics (PK) and pharmacodynamics (PD), particularly for drugs with narrow therapeutic indices.² Consequently, innovative quantitative strategies that enable individualized dose selection have gained increasing attention.

Among these strategies, physiologically based pharmacokinetic (PBPK) modeling has emerged as a powerful mechanistic framework for characterizing drug disposition and predicting systemic exposure in virtual populations.³ By integrating anatomical, physiological, and biochemical parameters, PBPK models allow simulation of drug concentration–time profiles not only at the population level but also at the individual level through the concept of a virtual twin (VT).^{3–5} In this approach, routinely available clinical data—such as age, body weight, renal and hepatic function tests, and concomitant medications—can be incorporated into individualized simulations to prospectively estimate drug exposure.⁴ This systems-based approach is particularly attractive for vulnerable populations in whom variability is pronounced, including geriatric patients. An early application of this approach is the work by Polasek et al., which demonstrated the feasibility of constructing individualized PBPK models to predict olanzapine exposure.⁶ Subsequent studies have extended the PBPK-VT framework to other drugs, including clozapine, apixaban, rivaroxaban, and caffeine, illustrating its ability to capture interindividual variability in PK and to support dose optimization.^{7–10}

Valproic acid (VPA) is a first-line antiepileptic drug (AED) and is an established effective treatment option for bipolar disorder across the acute and maintenance phases. It is used in geriatric patients as a therapeutic option for epilepsy, bipolar disorder, and comorbid neurological and psychiatric conditions.^{11,12} In clinical practice, VPA is typically initiated at a low dose (e.g., 250 mg/day) and gradually titrated based on clinical response and tolerability, following a “start low, go slow” approach.¹³ However, VPA has a narrow therapeutic index and is associated with potentially serious adverse effects, including gastrointestinal adverse effects, neurological symptoms (e.g., tremor, gait disturbance, and oversedation), hematologic abnormalities (e.g., thrombocytopenia), and hepatotoxicity, which warrant careful monitoring.^{14,15} Dose selection of VPA in elderly patients is especially challenging, as older adults represent a highly heterogeneous group, encompassing a wide age range with substantial variability in comorbidities and organ function.¹⁶ Clinical consensus suggests that older adults are generally more susceptible to adverse effects of AEDs.^{11,17,18} Cognitive impairment and mobility-related adverse effects are among the most frequently reported concerns in elderly patients, leading to high rates of treatment discontinuation.¹⁷

An additional challenge in the use of VPA specifically in geriatric patients is the limited availability of comprehensive PK data. Existing PK studies in elderly individuals are sparse. A study conducted in the 1980s reported significantly increased free VPA concentrations and reduced intrinsic clearance in elderly subjects compared with younger controls, suggesting age-related decreases in plasma protein binding and hepatic metabolic capacity.¹⁹ However, these findings were based

on a single oral dose administered to only six healthy elderly participants. Furthermore, the once-daily extended-release options now commonly used in clinical practice have primarily been studied in younger adults, with limited data available in geriatric populations.²⁰ As a result, dosing recommendations for elderly patients remain largely empirical and lack quantitative mechanistic justification.

Therapeutic drug monitoring (TDM) is often employed to guide VPA therapy and improve safety in clinical practice.²¹ The commonly accepted therapeutic range for total VPA plasma concentrations is 50–100 mg/L,²¹ which serves as a reference for assessing potential toxicity. While dose adjustments are primarily guided by clinical response and tolerability, TDM functions as a supportive tool, particularly in situations of suspected nonadherence or toxicity, given that several adverse effects are concentration-related.²² This approach is especially relevant in geriatric populations, where substantial interindividual PK variability is observed. Further, in elderly patients, TDM is useful for identifying issues such as nonadherence, which are common in older populations and can markedly influence drug exposure.²¹ Accordingly, TDM of VPA is frequently conducted in geriatric patients; however, it remains largely empirical and lacks a mechanistic basis for interpreting variability in drug exposure. An incomplete mechanistic understanding of how age-related physiological changes interact to influence VPA PK limits the ability to tailor dosing in geriatric patients. This limitation is particularly critical in this vulnerable population, who are at increased risk of exposure outside the therapeutic window and consequent toxicity.¹⁸

The PBPK-guided VT approach represents a promising strategy to address these challenges in geriatrics. By integrating patient-specific demographic and physiological characteristics into a mechanistic PK framework, this approach enables prospective estimation of individual VPA exposure in geriatric patients. This predictive capability may support the identification of an appropriate target maintenance dose to achieve clinically appropriate exposure while minimizing the risk of concentration-related toxicity and may complement conventional TDM by informing both dosing decisions and interpretation of monitoring results.

In the present study, we evaluated the ability of a PBPK-VT framework to predict VPA exposure in geriatric patients by accounting for age-related physiological variability and patient-specific clinical characteristics. Specifically, we aimed to¹ collect real-world PK data for valproate in geriatric patients and assess the predictive performance of PBPK-based exposure simulations, and² explore the feasibility of implementing VT modeling to support personalized VPA dosing in this vulnerable population. Ultimately, we propose a model-informed precision dosing strategy in which PBPK-based VT simulations guide optimal dose selection and complement TDM to refine therapy for geriatric patients.

METHODS

Clinical data

Study population. Clinical data were retrospectively collected from the electronic medical records of patients aged 65 years or older who visited the outpatient psychiatry clinic at Samsung Medical Center between March 2024 and August 2025. Eligible patients included those

diagnosed with bipolar and related disorders according to the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5), who had a documented history of VPA use—either divalproex sodium (DVS; Depakote® ER) or sodium valproate (NaVPA; Orfil® SR)—during the study period and available TDM data.

Demographic data (age, sex, height, weight, and body mass index) were extracted, along with laboratory parameters reflecting renal function (blood urea nitrogen, creatinine, estimated glomerular filtration rate [GFR], and electrolytes), hepatic function (aspartate aminotransferase [AST], alanine aminotransferase [ALT], alkaline phosphatase [ALP], total bilirubin, albumin, and prothrombin time), and hematologic profiles (white blood cell count, hematocrit, and platelet count). TDM data included measured serum drug concentrations and corresponding sampling time points.

This study was approved by the Institutional Review Board of Samsung Medical Center (IRB No. 2025-09-133) and conducted in accordance with the Declaration of Helsinki. All data were obtained from previously recorded clinical records without additional patient contact. Data were de-identified prior to analysis and handled in a secure research environment to ensure confidentiality and minimize risk to participants.

Statistical analysis. Descriptive statistical analyses were performed to summarize demographic, clinical, and laboratory characteristics of the study population. For exploratory visualization of trends in VPA trough concentrations (C_{trough}) according to covariates, C_{trough} was examined across clinically relevant subgroups using box-and-jitter plots. To minimize dose-related confounding arising from the non-linear PK of VPA, subgroup analyses were restricted to patients receiving a fixed daily dose of 500 mg. The statistical analyses and figure generation were conducted using R (version 4.5.2; R Foundation for Statistical Computing, Vienna, Austria).

PBPK model development

Simcyp® setting for population. The PBPK models were implemented using the Simcyp Simulator (version 25; Certara, Sheffield, UK). The built-in healthy volunteer and geriatric population libraries provided within Simcyp® were used to generate virtual populations for the simulations. For population-based trial simulations, we used a population size of 100, with 10 trials and 10 subjects per trial. For VT simulations, an individualized design was applied, consisting of 10 trials with one subject per trial.

PBPK model of VPA. The PBPK model for VPA was constructed and simulated in Simcyp Simulator (Version 25; Certara, Sheffield, UK). Among the available oral formulations of VPA, the patients included in this study received either DVS or NaVPA, both of which are extended-release (ER) formulations characterized by slower absorption compared with immediate-release (IR) products. Accordingly, an ER-specific VPA model was developed. The dosing interval for ER formulations was 24 hours, consistent with their prolonged elimination half-life. In addition, ER formulations produce 10–20% lower peak-trough fluctuations in serum concentrations compared with IR formulations administered more frequently.²³

The model was developed by integrating drug-specific physicochemical and PK parameters from the literature, including information from the previously published PBPK model of VPA,²³ particularly with respect to oral absorption parameters for ER formulations (Table S1). The previously reported model was primarily developed to predict drug–drug interactions.²³ However, when implemented in our simulations, the metabolism parameters specified for the ER formulation resulted in a slight overprediction of VPA exposure in both younger adults and our geriatric patients. Therefore, key physicochemical properties (SMILES, polar surface area, hydrogen bond donors and acceptors, and intrinsic solubility) were additionally incorporated, and metabolic parameters

(UGT-mediated enzyme kinetics, hepatic intrinsic clearance, and renal clearance) were modified based on literature data of VPA,^{23–25} while retaining several parameters from prior publications.

Drug-specific physicochemical properties were obtained from PubChem. The molecular weight input for the model corresponded to VPA (144.21 g/mol), as plasma concentrations in clinical studies are quantified in terms of the VPA moiety. Accordingly, when simulating dosing with NaVPA or DVS, dose corrections were applied to reflect VPA-equivalent amounts. Specifically, 538 mg of DVS corresponds to 500 mg of VPA, and 300 mg of NaVPA corresponds to 260 mg of VPA. VPA exhibits nonlinear, concentration-dependent protein binding,²⁶ which was incorporated into the model using a user-defined fraction unbound profile (7–15%) reflecting saturation of albumin binding sites.²⁷

Absorption of the ER formulation was described using the advanced dissolution, absorption, and metabolism (ADAM) model within Simcyp, incorporating a solubility of 1.3 mg/mL and Caco-2 permeability data.²⁸ The modified drug release profile for the monolithic ER system was incorporated as discrete percent-release values over time, based on published data.^{23,29} Drug distribution was described using a minimal PBPK informed by the prior model.²³ VPA elimination parameters were derived from published data, with glucuronidation—mediated primarily by UGT1A3, 1A4, 1A6, 1A8, 1A9, 1A10, and 2B7—representing the dominant metabolic pathway.^{23–25}

Validation of the VPA model. To validate the developed model, model performance was evaluated using datasets that were independent of those used for model construction, including published PK datasets^{30–33} and an in-house geriatric dataset.

Specifically, four single-dose studies and three multiple-dose studies of healthy adult volunteers were employed.^{30–33} For simulations, drug dose, administration schedule, age, and sex distributions were matched to the design of the corresponding clinical studies. For the studies where the age and sex distribution were not reported, default settings were applied, assuming an age range of 18–65 years and a 50% female distribution. Predictive performance was assessed by calculating the ratio of mean predicted to mean observed values for area under the concentration–time curve (AUC) and maximum plasma concentration (C_{max}). Model performance was considered acceptable if predicted AUC and C_{max} values were within a twofold range of the observed values, as previously described.^{34,35}

In addition, model validation was performed using observed PK data from our geriatric clinical sample to evaluate the applicability of the developed model in elderly patients. Validation was conducted by simulating steady-state C_{trough} according to the dose groups observed in the clinical sample (250, 260, 500, 520, 750, 780, 1,000, and 1,250 mg/day). Predictive performance was assessed by calculating the ratio of the mean predicted to mean observed C_{trough} values for each dose group.

Prediction of the PK of VPA in an elderly population

Following model validation, the developed PBPK model was applied to predict steady-state PK of VPA ER formulation in an elderly population (aged 65–98 years). These simulations were conducted to provide a population-level characterization of VPA PK, thereby complementing the limited availability of PK data in geriatric populations. Simulations were performed across a clinically relevant dose range of 250–1250 mg/day using the geriatric population model. Key PK parameters, including C_{max}, AUC, and C_{min}, were derived from the simulated concentration–time profiles.

Prediction of the PK of VPA in elderly individuals

A “virtual twin” profile for each study participant was generated by individualizing the Simcyp® geriatric population model according to participant-specific demographic and physiological characteristics. Simulations for each VT were performed using the same VPA dose

administered to the corresponding real patient. Steady-state C_{trough} for each VT was obtained by simulating 10 days of dosing with 10 trials per individual. Individualized input parameters were fixed across all 10 simulations, whereas the remaining non-fixed system parameters were sampled according to the predefined population variability. The mean predicted steady-state C_{trough} was then compared with the observed C_{trough} measured in patients under steady-state conditions. The observed C_{trough} values were obtained within 1 hour prior to the next scheduled dose, yet the exact timing of blood sampling relative to dosing was not recorded. To evaluate the impact of progressive covariate individualization on prediction performance, two levels of virtualization were implemented, based on previously reported approaches to VT modeling¹⁰: (i) VT Model I, incorporating demographic covariates (age, body weight, height, and sex), enabling subject-specific scaling of organ volumes and blood flows within the Simcyp simulator; and (ii) VT Model II, incorporating both demographic and physiological covariates (GFR, serum albumin concentration (g/L), and hematocrit (%)). These physiological covariates were fixed for each VT according to the measured values of the corresponding study participant, enabling further personalization of PK predictions.

VT PBPK model evaluation. To assess the performance of the VT models, predicted C_{trough} values were compared with matched observed concentrations for each patient. Predictive performance was quantified using the average fold error (AFE) and absolute average fold error (AAFE) calculated as follows:

$$AFE = 10^{\frac{1}{n} \sum \log\left(\frac{\text{Predicted C}_{\text{trough}}}{\text{Observed C}_{\text{trough}}}\right)} \quad (1)$$

$$AAFE = 10^{\frac{1}{n} \sum \left| \log\left(\frac{\text{Predicted C}_{\text{trough}}}{\text{Observed C}_{\text{trough}}}\right) \right|} \quad (2)$$

The AFE indicates the direction of prediction bias, reflecting whether the model systematically underestimates or overestimates observed concentrations, whereas the AAFE quantifies the magnitude of prediction error independent of direction. Model accuracy was primarily evaluated using AAFE, and validation of the VT model was considered acceptable if the predicted 95% confidence interval (CI) of the AAFE fell within the conventional bioequivalence range of 0.8–1.25.

Additionally, the proportion of predicted concentrations falling within 0.8- to 1.25-fold of the observed values was evaluated. As a more conservative assessment, the proportion of predictions within the narrower 0.9- to 1.11-fold range was also examined. For narrow therapeutic index (NTI) drugs—where relatively small differences in systemic exposure may lead to serious therapeutic failure or adverse drug reactions—a 20% deviation in exposure may be clinically relevant.³⁶ Therefore, a more conservative range of 90–111% has sometimes been considered as a bioequivalence margin for NTI drugs.^{36,37}

To further evaluate predictive performance, simple linear regression analysis was performed, and the coefficient of determination (R^2) was calculated to quantify the agreement between predicted and observed concentrations. In addition, Bland–Altman analyses were conducted to assess concordance between predicted and observed values. All statistical tests were two-tailed, and 95% CIs were computed. These statistical analyses were conducted using GraphPad Prism software, version 10.6.1 (GraphPad Software, San Diego, CA, USA).

RESULTS

Clinical data observation

A total of 114 patients were initially identified for the study. Among these, 15 patients were excluded due to noncompliance,

Table 1 Demographic and clinical characteristics of participants

Characteristics	Total
Demographics	
Age, years	73.97 ± 6.65
Female sex, <i>n</i> (%)	40 (54.10%)
Body weight, kg	63.21 ± 10.46
Height, cm	160.71 ± 8.30
Renal function	
Estimated GFR, mL/min/1.73 m ²	73.74 ± 15.67
Blood urea nitrogen, mg/dL	16.77 ± 5.24
Serum creatinine, mg/dL	0.91 ± 0.27
Hepatic function	
AST, U/L	26.00 (20.75–30.00)
ALT, U/L	18.50 (12.75–24.25)
ALP, U/L	64.00 (51.25–78.00)
Total bilirubin, mg/dL	0.40 (0.40–0.60)
Albumin, g/dL	4.11 ± 0.37
Total protein, g/dL	6.84 ± 0.67
Hematologic parameters	
Hematocrit, %	40.15 ± 4.84
White blood cell count, ×10 ⁹ /L	6.96 ± 3.63
VPA treatment	
Daily VPA dose, mg/day	500 (500–750)
Serum VPA concentration, µg/mL	45.19 (34.54–55.51)

Note: Values are presented as mean ± standard deviation or median (interquartile range), as appropriate. Abbreviations: ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; GFR, glomerular filtration rate; VPA, valproic acid.

20 patients were excluded because of incomplete anthropometric data (height and/or body weight), and 5 patients were excluded owing to insufficient laboratory data. Consequently, a total of 74 patients were included in the final analysis.

Clinical and demographic characteristics of the population can be found in Table 1. The mean age was 73.97 ± 6.65 years, and 40 patients (54.1%) were female. The median daily VPA dose was 500 (500–750) mg/day, and the median VPA C_{trough} was 45.19 (34.54–55.51) mg/L.

Box-and-jitter plots illustrating the influence of covariates on VPA trough concentrations following once-daily administration of 500 mg (*n* = 23) are presented in Figure 1. Although no statistically significant differences were observed due to the limited sample size, several descriptive trends were observed. Higher C_{trough} values were generally observed in older patients and in females. Increasing BMI tended to be associated with lower concentrations, whereas higher serum albumin levels (≥ 45 g/dL) were associated with higher C_{trough}. In contrast, no consistent trends were identified across categories of GFR or hematocrit.

PBPK model validation

The observed and simulated PK of VPA following single and multiple doses in healthy adult populations are summarized in

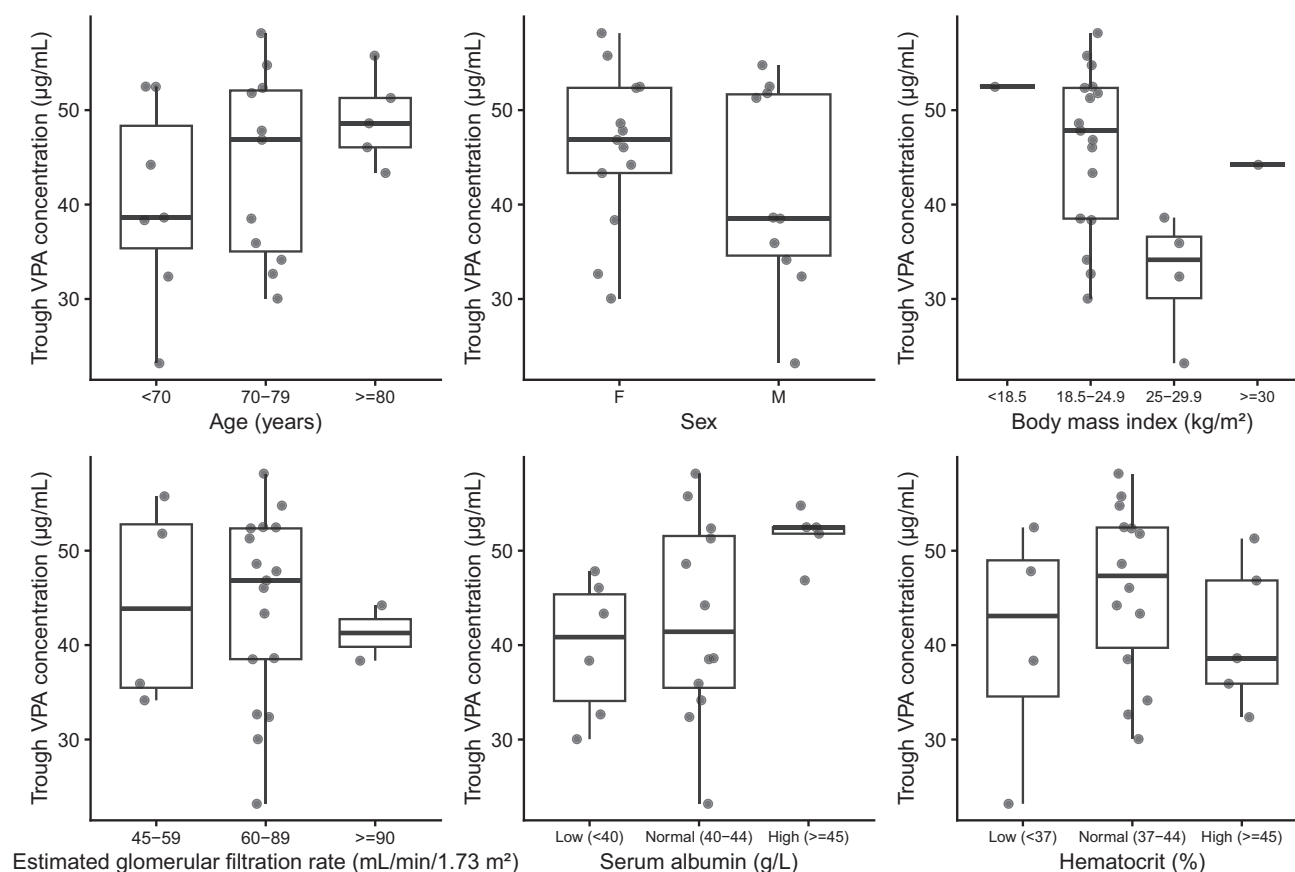


Figure 1 Distribution of trough valproic acid (VPA) concentrations across clinical covariates. Box-and-jitter plots illustrate trough VPA concentrations ($\mu\text{g/mL}$) in patients receiving 500mg/day ($n=23$), stratified by age, sex, body mass index, estimated glomerular filtration rate, serum albumin, and hematocrit. Boxes represent the interquartile range, horizontal lines indicate the median, and whiskers extend to 1.5 times the interquartile range. Individual data points are overlaid as jittered dots.

Table 2. Overall, the predicted C_{max} values were within 0.73 to 0.93 times the observed C_{max} values, and the predicted AUC values ranged from 0.88 to 1.15 times the observed AUC values, falling within the acceptable twofold limit.

Furthermore, simulated steady-state C_{trough} values in the geriatric population were compared with the corresponding observed C_{trough} values from our clinical sample for each dose group to further evaluate model performance in elderly patients (Table 3). Overall, the predicted C_{trough} values ranged from 0.92- to 1.37-fold of the observed values, falling within the acceptable twofold limit.

Prediction of the PK of VPA in an elderly population

Steady-state PK of VPA in the elderly population were simulated using the validated PBPK model across a dose range of 250–1250 mg/day (Table 4). Overall, simulated systemic exposure increased with escalating dose. However, the increase in AUC appeared less than dose-proportional at higher doses, consistent with the concentration-dependent protein binding and nonlinear PK of VPA. Following administration of 500 mg/day, the mean steady-state C_{min} (46.5 mg/L) was slightly below the lower limit of the commonly accepted therapeutic range (50 mg/L). In contrast,

mean C_{min} values fell within the therapeutic range when doses of 750, 1000, and 1250 mg/day were administered.

Following administration of 1000 mg/day in younger adults (18–65 years), the simulated steady-state AUC and C_{max} were 1508.8 mg·h/L and 72.7 mg/L, respectively. In comparison, exposure in older adults was higher, with AUC and C_{max} values increased by 1.35- and 1.30-fold, respectively, relative to younger adults at steady state. These differences exceed the conventional bioequivalence limits (80–125%) and may therefore be clinically meaningful, particularly for NTI drugs.

VT PBPK model evaluation

The predictive performance of the individualized VT models is summarized in Table 5. Incorporation of demographic covariates (VT Model I) demonstrated acceptable predictive performance, yielding an AFE of 1.02 (95% CI, 0.98–1.07) and an AAFE of 1.16 (1.13–1.20), with 79.7% and 41.3% of predictions within the 0.8–1.25- and 0.9–1.11-fold ranges, respectively. Further individualization through inclusion of physiological parameters (VT Model II) provided improved performance (AFE of 1.00 [0.97–1.04]; AAFE of 1.13 [1.10–1.16]), with 87.8% and 50.7% of predictions meeting the respective criteria.

Table 2 Comparison of observed and simulated pharmacokinetic parameters for model validation in healthy adult populations

PK study			PK parameters			
Trial	Dose		Observed		Simulated	
			AUC ^a		AUC ^{as}	
			C _{max} (ng/mL)	(ng·h/mL)	C _{max} (ng/mL)	(ng·h/mL)
Dutta et al., 2004a ³⁰ (18–50 years, f=0.375, fed)	500 mg	Mean	29.7	1056	26.9	1007
	Single dose	SD	27	33	6.8	299
		Simulated/observed			0.91	0.95
Dutta et al., 2004b ³¹ (18–55 years, f=0.128, fasted)	500 mg	Mean	25.3	831	24.9	959
	Single dose	SD	8.95	295	6.1	296
		Simulated/observed			0.98	1.15
Dutta et al., 2004b ³¹ (18–55 years, f=0.128, fed)	500 mg	Mean	28.3	898	26.0	981
	Single dose	SD	7.63	295	6.3	286
		Simulated/observed			0.92	1.09
Qiu et al., 2003 ³³ (18–65 years, f=0.5, fed)	500 mg	Mean	29.6	967	27.6	1037
	Single dose	SD	6.7	371.3	7.1	313
		Simulated/observed			0.93	1.07
Dutta et al., 2004c ³² (19–55 years, f=0.125, fed)	1000 mg	Mean	84.8	1728	68.8	1437
	Steady-state	SD	8.7	216	11.7	256
		Simulated/observed			0.81	0.83
Qiu et al., 2003 ³³ (18–65 years, f=0.5, fasted)	1000 mg	Mean	80.5	1592	72.6	1472
	Steady-state	SD	18.6	402	15.2	319
		Simulated/observed			0.90	0.92
Qiu et al., 2003 ³³ (18–65 years, f=0.5, fed)	1000 mg	Mean	99.4	1709	72.7	1509
	Steady-state	SD	15.7	276	13.7	307
		Simulated/observed			0.73	0.88

Abbreviations: AUC, area under the curve; C_{max}, maximum plasma concentration; PK, pharmacokinetic; SD, standard deviation. ^aAUC (0–∞) for a single dose, AUC (0–24 h) for steady-state.

Figure 2 presents predicted versus observed trough concentrations for the full clinical sample ($n=74$) across VT models with increasing levels of covariate virtualization (**Figure 2a,b**), with corresponding Bland–Altman plots shown in **Figure 2c,d**. Inclusion of physiological parameters improved the alignment between the predicted and observed C_{trough} values, with the R^2 increasing from 0.7823 in VT Model I to 0.8742 in VT model II. Bland–Altman analyses showed minimal systematic bias in both models, with reduced variability after inclusion of physiological parameters (SD of bias: 15.90 vs. 12.56 mg/L). Overall, progressive incorporation of patient-specific covariates improved predictive accuracy and reduced bias.

DISCUSSION

The present study evaluated the predictive performance of the PBPK-guided VT framework for individualized exposure prediction in a real-world dataset of geriatric patients receiving VPA. By implementing PBPK models incorporating virtual geriatric populations and accounting for patient-specific demographic and physiological characteristics, the models demonstrated reasonable accuracy and precision in predicting VPA concentrations. While

this approach has increasingly been recognized as a promising strategy for model-informed precision dosing,³⁸ its performance has rarely been evaluated in routine clinical settings using real-world TDM data. In particular, evidence supporting PBPK-guided personalization in elderly patients—who represent a population with substantial physiological variability and high clinical need—remains limited. The present study, therefore, provides clinically relevant validation of the VT approach within a geriatric hospital setting and contributes to bridging the gap between mechanistic PBPK modeling and real-world therapeutic decision-making.

The PBPK-VT model of VPA may complement clinical decision-making by informing individualized dosing and enhancing the interpretation of TDM data. To support individualized dosing, patient-specific demographic and physiological information is incorporated into the model to generate individualized predictions of VPA exposure across a range of dosing scenarios. These simulations may be used to estimate a target maintenance dose expected to achieve concentration within the therapeutic range while minimizing the risk of concentration-related toxicity. In addition, to complement TDM data, observed VPA concentrations can be compared with model-predicted exposure. Discrepancies

Table 3 Comparison of observed and simulated steady-state Ctrough for model validation in geriatric patients in the clinical study (n = 74)

VPA Dose		Ctrough (mg/L)	
		Observed	Simulated
250 mg/day n = 8	Mean	22.9	27.3
	SD	7.2	8.1
	Simulated/observed		1.19
260 mg/day n = 7	Mean	24.2	28.2
	SD	4.1	8.3
	Simulated/observed		1.17
500 mg/day n = 23	Mean	44.0	46.5
	SD	9.5	11.3
	Simulated/observed		1.06
520 mg/day n = 6	Mean	41.3	47.7
	SD	4.9	11.5
	Simulated/observed		1.15
750 mg/day n = 18	Mean	55.0	59.2
	SD	11.0	12.9
	Simulated/observed		1.08
780 mg/day n = 1	Mean	44.2	60.5
	SD	-	13.2
	Simulated/observed		1.37
1000 mg/day n = 10	Mean	68.6	69.8
	SD	10.8	16.7
	Simulated/observed		1.02
1250 mg/day n = 1	Mean	88.0	81.2
	SD	-	23.4
	Simulated/observed		0.92

Table 4 Simulated pharmacokinetic parameter values for valproic acid at steady-state in the elderly population (aged 65–98 years) using the PBPK model

VPA Dose	PK parameters (mean ± SD)		
	C _{max} (mg/L)	AUC (0–24 h) (mg·h/L)	C _{min} (mg/L)
250 mg/day	33.9 ± 8.5	755.8 ± 197.8	27.3 ± 8.1
500 mg/day	59.2 ± 11.8	1305.8 ± 270.7	46.5 ± 11.8
750 mg/day	77.8 ± 13.6	1693.8 ± 303.7	59.2 ± 12.9
1000 mg/day	94.2 ± 18.7	2030.2 ± 416.5	69.8 ± 16.7
1250 mg/day	111.9 ± 26.4	2396.3 ± 597.8	81.2 ± 23.4

Note: Data are presented as mean ± standard deviation. Abbreviations: AUC, area under the curve; C_{max}, maximum plasma concentration; C_{min}, minimum plasma concentration; PK, pharmacokinetic; SD, standard deviation.

between predicted and observed concentrations may provide mechanistic insight into the underlying causes, such as altered PK or nonadherence, thereby improving the interpretation of TDM results. Overall, PBPK-VT is positioned to support individualized

target dose selection and the interpretation of monitoring data by providing prospective, mechanistic estimates of drug exposure. Importantly, the predictive performance of the PBPK model was evaluated not only against PK data obtained from younger healthy volunteers but also against independent PK data derived from our geriatric clinical sample. At the population level, predicted Ctrough values generated using the virtual geriatric population ranged from 0.92- to 1.37-fold of the observed concentrations across dose groups in our clinical data, indicating overall good agreement between simulations and clinical data. This indicates that the model reliably characterizes geriatric data at the population level and may help compensate for the limited availability of clinical PK data of VPA in elderly populations.

AAFE (95% CI) for the VT model I integrating demographic covariates met the predefined bioequivalence criteria with 1.16 (1.13–1.20). In addition, nearly 80% of predicted concentrations fell within the 0.8- to 1.25-fold range of the observed values. These findings suggest that even when only demographic information is available, incorporation of such covariates into the VT model can provide reasonably reliable predictions of Ctrough. This observation is consistent with previous population pharmacokinetic (PopPK) analyses, which have identified body weight, sex, and age as major covariates influencing VPA exposure.¹⁵ Increased VPA clearance with higher body weight has been reported, reflecting the association between body size and organ capacity for drug elimination.^{39–43} Sex-related differences have also been reported, with lower clearance of VPA observed in females,^{15,39–41} potentially attributable to reduced UDP-glucuronosyltransferase activity.⁴⁴ Although age has been identified as an influential covariate in prior studies, most investigations focused on pediatric populations,^{45–47} where developmental changes in drug clearance predominate and are not directly applicable to age-related physiological changes observed in older adults. Aging in geriatric patients is associated with gradual changes in hepatic function and protein binding that may influence PK through distinct mechanisms.⁴⁸ In our geriatric clinical sample, exploratory analysis within the 500 mg/day group (n = 23) suggested a trend toward higher VPA concentrations with increasing age.

In the present study, VT Model II, which incorporated both demographic data and physiological parameters, including serum albumin, hematocrit, and GFR, demonstrated the best predictive performance for VPA concentration. Nearly 90% of patients had predicted trough concentrations within the conventional bioequivalence range (0.8- to 1.25-fold), whereas 50.7% met the more stringent 0.9- to 1.11-fold criterion. This indicates that a substantial proportion of predictions, approximately half of the cases, remained outside strict NTI bounds, highlighting that it does not fully eliminate uncertainty in individual exposure estimates and should therefore be applied as a complementary tool to TDM in clinical settings. Among the incorporated covariates, serum albumin concentration likely contributed most to the improvement in model predictability. Exploratory analysis in the geriatric patients suggested a trend toward higher VPA concentrations at serum albumin levels higher than 45 g/L. This may be explained by the high protein-binding properties of VPA (approximately 90–95%),¹⁵ whereby increased albumin levels enhance protein binding, reducing the unbound fraction. This consequently decreases the fraction available for clearance, resulting in higher total VPA concentrations.

Table 5 Predictive performance of the VT models for VPA: AFE, AAFE and proportion of predictions within bioequivalence ranges for individual trough concentrations

Model	AFE (95% CI)	AAFE (95% CI)	% in 0.8- to 1.25-fold range	% in 0.9- to 1.11-fold range
VT Model I	1.02 (0.98–1.07)	1.16 (1.13–1.20)	79.7%	41.3%
VT Model II	1.00 (0.97–1.04)	1.13 (1.10–1.16)	87.8%	50.7%

Abbreviations: AAFE, absolute average fold error; AFE, average fold error.

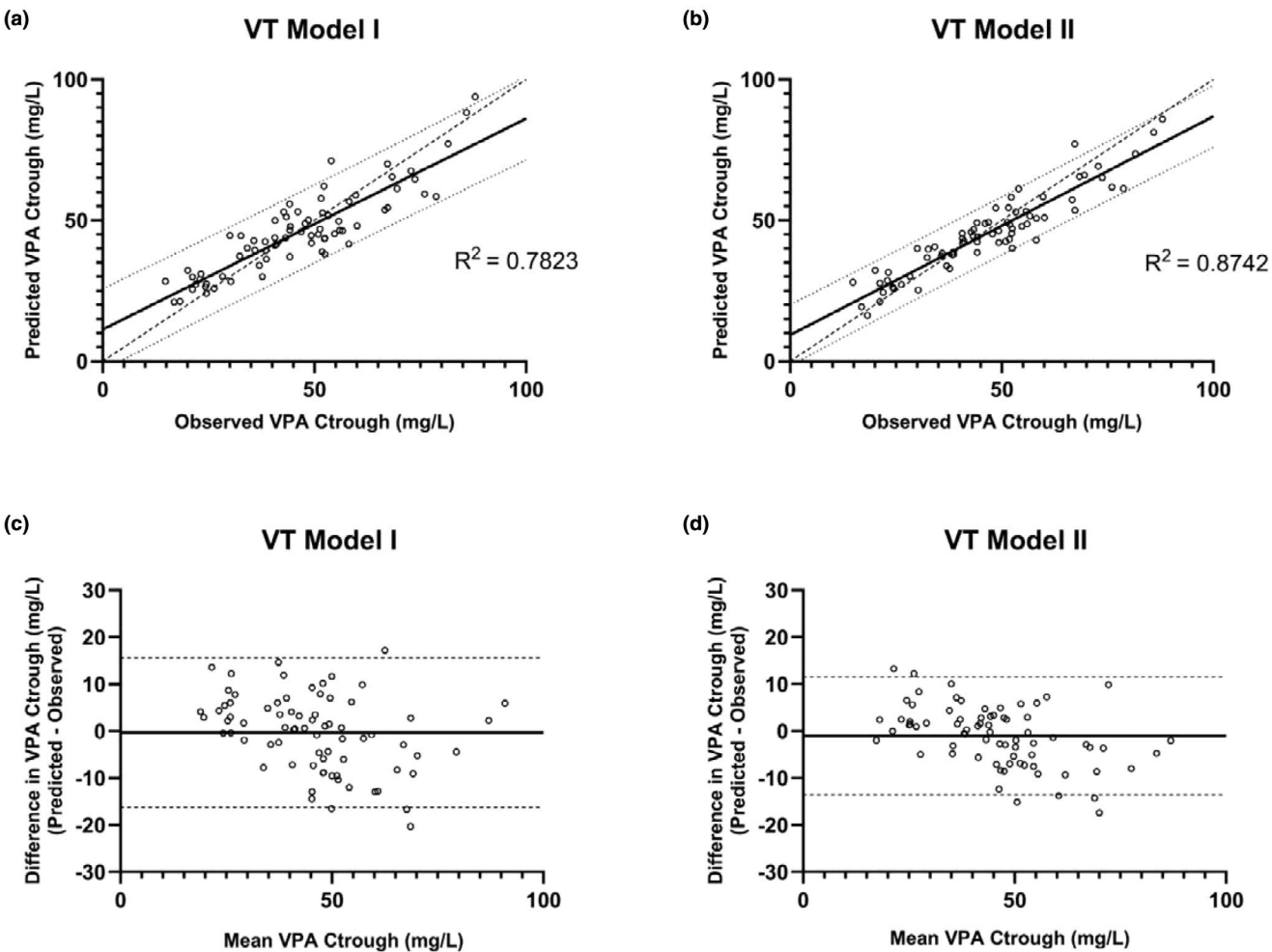


Figure 2 Correlations between predicted and observed steady state trough valproic acid (VPA) concentrations using virtual twin (VT) models in the full clinical sample ($n = 74$) (a) Predicted versus observed VPA trough concentrations for VT Model I, incorporating demographic covariates; (b) Predicted versus observed VPA trough concentrations for VT Model II, incorporating both demographic and physiological parameters; (c and d) Corresponding Bland–Altman plots for VT Model I and Model II. In panels (a) and (b), the solid black line represents the linear regression fit, the small dotted lines represent the 95% prediction intervals, and the larger dotted lines represent the line of unity. In panels (c) and (d), the solid line represents the mean difference between predicted and observed VPA trough concentrations, while the dotted lines represent the 95% confidence interval.

Importantly, although increased albumin levels may elevate total VPA concentrations, they do not necessarily increase the pharmacologically active unbound concentration.⁴⁹ The unbound concentration may remain relatively stable due to a compensatory reduction in clearance,^{50,51} suggesting that albumin-driven changes in total VPA levels may not directly translate into increased toxicity risk. In contrast, renal elimination accounts for only 1–3% of VPA clearance,²³ indicating a limited contribution of GFR to exposure prediction.

With respect to hepatic function, while we collected clinical variables reflecting hepatic function, including AST, ALT, ALP, and total bilirubin (Table 1), these variables could not be incorporated into the VT model due to the absence of this capability within the current model. Given that VPA is mainly metabolized by the liver,¹⁵ inclusion of these variables may further improve the model performance.

Genetic information related to CYP or UGT enzymes was not incorporated into the VT simulations; nevertheless, adequate predictive

performance was achieved. VPA metabolism occurs primarily via glucuronidation, whereas mitochondrial β -oxidation and cytochrome P450-mediated oxidation contribute to a lesser extent.²⁴ Glucuronidation is mediated by multiple UGT isoforms, including UGT1A3, UGT1A4, UGT1A6, UGT1A8, UGT1A9, UGT1A10, and UGT2B7,^{24,52} with UGT1A6, UGT1A9, and UGT2B7 accounting for approximately 50% of VPA metabolism.⁵³ Overall, the major metabolic pathways of VPA are not considered highly sensitive to genetic polymorphism. Although certain variants, such as UGT1A1 polymorphisms, have been reported, they are not major determinants of VPA clearance.^{24,52} Furthermore, CYP-mediated oxidation accounts for less than 10% of total VPA metabolism, which is mainly mediated via CYP2C9, CYP2A6, CYP2B6, and possibly CYP2C19.⁵⁴ Consequently, omission of CYP or UGT genotype information was unlikely to substantially influence the prediction of VPA trough concentrations in the present study. As genotyping of these metabolic enzymes is not routinely performed in clinical practice, this may enhance the practical applicability of the VT model in real-world clinical settings.

Our study has several limitations. First, the PBPK model was developed specifically for ER formulations, as all patients in our clinical sample received ER VPA administered once daily. Therefore, application of the VT framework to IR formulations would require development and independent verification of an IR-specific model. Second, concomitant medications and individualized genetic information related to UGT or CYP enzymes were not incorporated due to the absence of such data in our clinical setting. Although incorporation of genetic information might further refine prediction accuracy, the current model nevertheless demonstrated satisfactory predictive performance, meeting predefined bioequivalence criteria. Third, the study was conducted in a single-center geriatric population with bipolar disorder, with a relatively limited sample size, which may restrict the generalizability of the findings to broader elderly populations with different clinical characteristics or ethnic backgrounds. Fourth, only C_{trough} obtained from routine TDM were available for validation; richer PK sampling would have enabled evaluation of additional exposure metrics.

Despite these limitations, this study contributes to the emerging field of PBPK-VT modeling, for which only a limited number of applications have been reported to date. While prior studies have demonstrated the feasibility of this approach across selected compounds, the present work extends its application to VPA, a drug characterized by substantial PK variability and a narrow therapeutic index. In addition, our study supports the application of the VT framework in special populations, particularly elderly patients, who exhibit substantial physiological variability and increased vulnerability to adverse effects, thereby highlighting the clinical need and potential utility of VT-based approaches in special populations. Collectively, these findings further support the potential of PBPK-VT as a tool for precision dosing.

In conclusion, the present study verified the ability of a PBPK-based VT framework to predict individual VPA plasma concentrations measured through TDM in geriatric patients. This VT approach may serve as a foundation for a clinically applicable prescribing support tool, enabling clinicians to prospectively assess the risk of underexposure or overdose. By incorporating patient demographics and routinely available physiological parameters,

including serum albumin concentration, the model provides a practical strategy to support more informed drug selection and dose optimization in elderly patients receiving VPA.

SUPPORTING INFORMATION

Supplementary information accompanies this paper on the *Clinical Pharmacology & Therapeutics* website (www.cpt-journal.com).

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CONFLICT OF INTEREST

The authors declared no competing interests for this work.

AUTHOR CONTRIBUTIONS

Y.J.J. and E.H. wrote the manuscript; Y.J.J., E.H. designed the research; Y.J.J., D.G.H., and E.H. performed the research; Y.J.J., D.G.H., and E.H. analyzed the data.

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